CHAPTER 5

Introducing Bitcoin

From Digital Cash Concepts to Decentralized Currency

TC789 – Teknologi Blockchain

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SOURCE

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AUDIENCE

Undergraduate
Computer Science

Learning Objectives

By the end of this lecture, students will be able to:

1. Describe Bitcoin's historical origins and the core problems it solves
2. Explain the roles of private keys, public keys, and Bitcoin addresses
3. Describe the structure and full life cycle of a Bitcoin transaction
4. Explain how the Script language enforces transaction spending conditions
5. Describe the structure of a Bitcoin block and the blockchain as a whole
6. Explain Proof of Work and the step-by-step mining process
7. Trace the evolution of mining hardware from CPU to ASIC
8. Identify Bitcoin's key benefits and practical limitations

Chapter Overview

1

Pre-Bitcoin History

Hashcash · B-money · BitGold

2

Bitcoin Basics

Definition · components · bird's-eye view

3

Keys & Addresses

Private key · public key · address types

4

Transactions

UTXO · life cycle · script language

5

Blockchain Structure

Block · header · genesis · forks

6

Mining

PoW · hardware evolution · pools

The Road to Bitcoin

Decades of research in cryptography and digital cash converged in Bitcoin

1982	**David Chaum**	Blind signatures for untraceable digital payments	*Central bank required*
1997	**Adam Back**	Hashcash — computational puzzle; anti-spam foundation of PoW	*No currency, no ledger*
1998	**Wei Dai / Nick Szabo**	B-money & BitGold — PoW-based money concepts, adjustable difficulty	*Lacked consensus mechanism*
2004	**Hal Finney**	Reusable Proof of Work (RPoW) — first person to receive Bitcoin	*Still centralized DB*
2008	**Satoshi Nakamoto**	Bitcoin: A Peer-to-Peer Electronic Cash System — no trusted third party	*Problem solved*

Common weakness of all pre-Bitcoin schemes: they relied on a central trusted server — Bitcoin eliminated this.

What Is Bitcoin?

Definition

- Bitcoin (capital B) = the protocol, platform, and peer-to-peer network
- bitcoin (lowercase b) = the digital currency unit
- Introduced in 2008 by Satoshi Nakamoto (identity unknown — pseudonym)
- First fully decentralized digital currency — no central authority required
- Solves the double-spend problem via distributed blockchain + Proof of Work

Key Facts

Launch Year
2008 / 2009

Max Supply
21 Million BTC

Smallest Unit
1 Satoshi = 0.00000001 BTC

Block Time
~10 minutes

Bitcoin: A Bird's-Eye View

Core Components

- Digital Keys
- Addresses
- Transactions
- Blockchain
- Miners
- Network Nodes
- Wallets (software)

Payment Flow (User Perspective)

1. **Receiver shares:** Bitcoin address or QR code
2. **Sender signs:** Transaction signed with private key & broadcast
3. **Mempool:** Miners select transactions by fee priority
4. **Mining:** Miners verify, build block, solve PoW puzzle
5. **Block broadcast:** All nodes verify and update their chain
6. **Confirmed:** 3–6 block confirmations = final settlement

Digital Keys and Addresses

Private Key	Public Key	Bitcoin Address
256-bit random number (secp256k1 curve range)	Mathematically derived from private key via ECC	Hash of public key: SHA-256 then RIPEMD-160
Digitally signs transactions — proves ownership	Derivation is one-way — cannot reverse to get private key	Base58Check encoded → 26–35 readable characters
Stored in Wallet Import Format (WIF)	Visible on the blockchain; used to verify signatures	Begins with 1 (P2PKH) or 3 (P2SH)
Must stay secret — loss is permanent and irrecoverable	Compressed (33 bytes) or uncompressed (65 bytes)	Often shared as QR code — do not reuse addresses
Mini key format (≤30 chars) for compact storage	Standard since Bitcoin Core v0.6: compressed format	Reusing addresses reduces privacy and security

Private Key ——→ Public Key ——→ Bitcoin Address (one-way mathematical derivation)

Transactions & the UTXO Model

UTXO = Unspent Transaction Output

- Bitcoin has no balances — only unspent coins (UTXOs)
- Each input spends a previous UTXO
- Each output creates new UTXOs for recipients
- Coins measured in Satoshis (1 BTC = 100,000,000 Satoshis)
- Analogy: pay $50 for $30 item — $20 change is a new UTXO
- Transactions are public and visible on blockchain explorers

Transaction Data Structure

Field	Size	Description
Version	4 bytes	Transaction processing rules
Input Counter	1–9 bytes	Number of inputs
Inputs	Variable	UTXO references + unlocking scripts (ScriptSig)
Output Counter	1–9 bytes	Number of outputs
Outputs	Variable	Satoshi amounts + locking scripts (ScriptPubKey)
Lock Time	4 bytes	Earliest valid time or block height

Transaction Life Cycle

From initiation to final confirmation on the Bitcoin network

1 Sign	2 Broadcast	3 Mempool	4 Mining	5 Block Found	6 Confirmed
Wallet signs the transaction with sender's private key	Serialized transaction sent to the network via flooding	Enters memory pool; miners select by fee priority	Miners verify, build block, and solve PoW puzzle	Mined block broadcast; all nodes verify and add to chain	3–6 block confirmations = final, irreversible settlement

Fee = sum(inputs) − sum(outputs) | Higher fee → miners pick up transaction faster | Confirmation time: 10 min to 12+ hours

Transaction Malleability (changing TX ID before confirmation) was fixed by the SegWit soft fork upgrade.

The Bitcoin Script Language

Script Basics

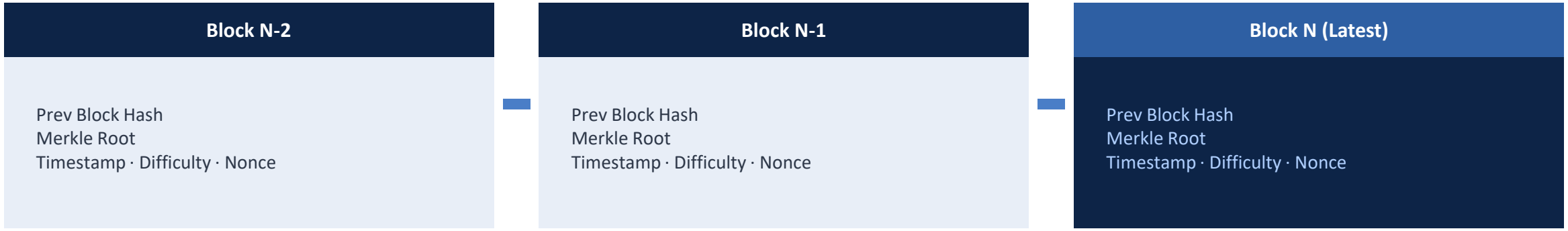
- Stack-based language (Forth-like, Reverse Polish Notation)
- NOT Turing-complete — no loops to prevent infinite execution
- ScriptPubKey = locking script (conditions to spend output)
- ScriptSig = unlocking script (proof of key ownership)
- Evaluated by concatenating ScriptSig + ScriptPubKey on stack
- Contracts possible: escrow, multisig, time-locked payments

Standard Transaction Types

P2PKH	Pay to Public Key Hash — most common; sends to a bitcoin address (starts with 1)
P2SH	Pay to Script Hash — sends to a script hash; addresses start with 3
MultiSig	M-of-N — requires M signatures from N keys; used for escrow and shared wallets
Coinbase	First transaction in every block; creates new bitcoins (block reward for miner)
OP_RETURN	Stores arbitrary data (≤40 bytes) on-chain; output is provably unspendable

Blockchain Structure in Bitcoin

A public ledger: timestamped, ordered, and immutable record of all transactions



Block Header Fields

Field	Size	Description
Version	4 bytes	Block validation rules version
Previous Block Hash	32 bytes	Double SHA-256 of prior header; changing any block invalidates all subsequent hashes
Merkle Root	32 bytes	Hash of all transactions — alters if any transaction changes
Timestamp	4 bytes	Approximate block creation time (Unix epoch format)
Difficulty Target	4 bytes	Current PoW target; adjusted every 2016 blocks to maintain 10-min block time
Nonce	4 bytes	32-bit number miners increment until hash < difficulty target

Genesis Block and Protocol Forks

The Genesis Block

"The Times 03/Jan/2009
Chancellor on brink of second bailout for banks"

- First block in the Bitcoin blockchain — hardcoded in Bitcoin Core source
- Embedded timestamp proves this block was not pre-mined
- Message hints at Satoshi's motivation: distrust of central banks
- Block height 0 — all subsequent blocks trace back to this block

Blockchain Forks

Soft Fork	Backward-compatible upgrade. Old nodes still work. Only miners must update. Example: SegWit.
Hard Fork	Breaking change. All nodes must upgrade. Can create a permanent new chain. Example: Bitcoin Cash.
Orphan Block	Accepted then discarded when a longer valid chain was built without it.
Stale Block	Solved block made irrelevant when another miner found a block at the same time.

51% Attack: If a pool controls >51% of network hash rate, it can potentially double-spend or rewrite recent transaction history. The network always follows the longest valid chain.

Bitcoin Mining

What Mining Does

- Validates and confirms transactions in new blocks
- Generates new bitcoins (the block reward)
- Secures the network against fraud and double spending
- Maintains a consistent ~10-minute block interval

Proof of Work Requirement

$$H(N \,||\, P_hash \,||\, Tx\ldots) < Target$$

N = nonce | P_hash = previous block hash | Only solvable by brute force

Mining Algorithm

1. Retrieve previous block's header from the network
2. Assemble validated pending transactions into a proposed block
3. Compute double SHA-256(prev_header || nonce || block)
4. If hash < difficulty target → solved → broadcast block & earn reward
5. Else → increment nonce and repeat from step 3

Difficulty adjusted every 2016 blocks (~2 weeks) to maintain ~10-minute block time

Mining Rewards & Hardware Evolution

Block Reward Halving

2009	**50 BTC**	*Bitcoin launched*
2012	**25 BTC**	*First halving*
2016	**12.5 BTC**	*Second halving*
2020	**6.25 BTC**	*Third halving*
~2140	**0 BTC**	*Only fees remain; 21M cap reached*

Halves every 210,000 blocks (~4 years) | Total supply capped at 21,000,000 BTC

Mining Hardware Evolution

CPU 2009–2010	Original Bitcoin client. Laptop/desktop. Quickly became unprofitable.
GPU 2010–2012	Graphics cards; parallel SHA-256 via OpenCL. Overclocking common.
FPGA 2012–2013	Programmable hardware (Verilog/VHDL); faster than GPU; required expertise.
ASIC 2013–now	Purpose-built SHA-256 chips. Dominant today. Industrial mining centers required.

Mining Pools: groups sharing hash power & rewards. Risk: >51% control = 51% attack risk.

Key Terms & Definitions

UTXO	Unspent Transaction Output — the fundamental unit of Bitcoin value; coins are UTXOs, not balances
Nonce	A 32-bit number that miners repeatedly change to produce a hash below the current difficulty target
P2PKH	Pay-to-Public-Key-Hash — most common Bitcoin transaction type; sends coins to a bitcoin address
Soft Fork	Backward-compatible protocol change; old nodes remain functional on the network
Hash Rate	Speed at which the network computes hashes; measured in TH/s or EH/s (exa-hashes/second)
PoW	Proof of Work — miners solve a hard hash puzzle to earn the right to add a block to the chain
Mempool	Memory pool — holds unconfirmed transactions waiting to be included in a mined block
Coinbase	First transaction in every block; created by the miner to claim the block reward and fees
Hard Fork	Breaking protocol change; requires all nodes to upgrade; can create a permanent new chain
WIF	Wallet Import Format — compact, human-readable encoding of a Bitcoin private key for portability

Benefits vs. Limitations

	Benefits
✓	Fully decentralized — no central authority required
✓	Double-spend problem solved via PoW + blockchain
✓	Transparent — all transactions publicly auditable
✓	Pseudonymous — addresses not directly tied to real identity
✓	Finite supply (21M BTC) — built-in inflation resistance
✓	Permissionless — anyone can participate globally

	Limitations
!	High energy consumption (PoW mining is computationally intensive)
!	Slow confirmation — 10 minutes to hours per transaction
!	Volatile transaction fees, especially during high demand
!	Scalability bottleneck — limited block size and throughput
!	Transaction malleability (partially mitigated by SegWit)
!	Mining centralization risk — 51% attack if pools grow too large
!	Regulatory uncertainty and varying legal treatment globally

Key Takeaways

1. Bitcoin (2008) was the first decentralized digital currency, built on 30+ years of cryptographic research
2. It solves the double-spend problem without any central authority via blockchain + Proof of Work
3. Private keys sign transactions; public keys verify; addresses are public-key hashes (Base58Check)
4. The UTXO model tracks ownership: each input spends a previous output; no traditional "balances"
5. Mining adds blocks by solving PoW, generates new BTC, and secures the network against tampering
6. Block reward halves every ~4 years (210,000 blocks); total supply permanently capped at 21 million BTC
7. Mining hardware evolved: CPU → GPU → FPGA → ASIC; solo mining is now largely uneconomical
8. SegWit fixed transaction malleability; soft forks are backward-compatible; hard forks require all upgrades

Discussion & Q&A

Chapter 5 — Introducing Bitcoin

Q1	How does Bitcoin solve the double-spend problem without a bank or central authority?
Q2	Why is Proof of Work energy-intensive by design — is this waste or a security feature?
Q3	What would happen if a mining pool obtained more than 51% of the network hash rate?
Q4	How does the reward halving mechanism affect the incentive to mine over time?

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